$$u\cos\theta = c\,\frac{\lambda - \lambda_0}{\lambda_0}\ ; \qquad (11)$$

Substituting the data gives

$$u\cos\theta = 5.85 \times 10^3\ \mathrm{m \cdot s^{-1}}\ . \qquad (12)$$

From (12), for θ=30°, a sodium atom has speed

$$u_1 = 6.76 \times 10^3\ \mathrm{m \cdot s^{-1}}\ ;$$

For $\theta = 45°$, the sodium atom has speed

$$u_2 = 8.28 \times 10^3\ \mathrm{m \cdot s^{-1}}\ .$$

For motion at angles to z in 30° -45°, resonant sodium atoms have speeds in the range

$$6.76 \times 10^3\ \mathrm{m \cdot s^{-1}} < u < 8.28 \times 10^3\ \mathrm{m \cdot s^{-1}}\ . \qquad (13)$$

From (3), the sodium atom's velocity change in resonant absorption is

$$u - V = \frac{h}{M\lambda} e_z\ , \qquad (14)$$

The magnitude of this velocity change is

$$|u - V| = \frac{h}{M\lambda}\ ; \qquad (15)$$

Substituting the data gives

$$|u - V| = 2.9 \times 10^{-2}\ \mathrm{m \cdot s^{-1}}\ . \qquad (16)$$